CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 10: **Light: Shadows and Reflections** | Quick revision, one page

Light and shadows

- Light travels in straight lines. Luminous objects give out their own light (Sun, stars, bulb); non-luminous ones reflect it (Moon, book).
- Transparent: light passes fully (clear glass). Translucent: partly (butter paper). Opaque: not at all (wood, metal).
- Shadow: needs a source of light, an opaque object and a screen. Shadow of a closer object is larger and darker.
- Umbra: the darkest part of a shadow; penumbra: the partly dark region around it.

Reflection of light

- Laws: angle of incidence = angle of reflection; incident ray, reflected ray and normal are in the same plane.
- Regular reflection: from smooth surfaces (mirrors); diffuse (irregular) reflection: from rough surfaces.

Image in a plane mirror

- Image is virtual (cannot be got on a screen), erect and of the same size as the object.
- Image distance behind the mirror = object distance in front of it.
- Lateral inversion: left appears right and right appears left (AMBULANCE).

Pinhole camera and applications

- Pinhole camera forms an inverted real image; a larger hole gives a brighter but blurred image.
- Periscope uses two plane mirrors; kaleidoscope uses mirrors to form patterns by multiple reflection.

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